Shape-Optimization-Stokes-Darcy-System

1 Abstract

We propose a novel shape optimization model governed by the Stokes–Darcy system with the Beavers–Joseph–Saffman interface conditions, aiming to minimize energy dissipation within the free-flow region. By performing shape sensitivity analysis that accounts for the effects arising from the nonhomogeneous interface conditions and multiphysics coupling, we derive both distributed and interface-type Eulerian derivatives. To numerically solve this multiphysics coupling PDE-constrained optimization problem, we propose a shape gradient flow-based algorithm, which is discretized via the finite element method and realized with the moving body-fitted grid technique. Numerical examples demonstrate that the proposed algorithm enables shape optimization simulations under diverse permeability and viscosity coefficients, with effectiveness in both 2D and 3D configurations.

Code Package Structure

The code package is organized as follows:

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Shape-Optimization-Stokes-Darcy-System/
|
|-- Sec 4.1: Single-interface around a porous region/
| |-- Case 4.1.1/
| |-- Case 4.1.2/
|
|-- Sec 4.2: Multi-interface optimization in an L-shaped configuration/
|
|-- Sec 4.3: Diffuser model/
| |-- 2D/
| |-- 3D/
|
|-- Sec 4.4: Embedded vessel-tissue coupling model/
| |-- 2D/
| |-- 3D/
```

2 Common Numerical Settings

All codes are implemented in FreeFEM++. The free-flow region Ω_f is governed by the Stokes equations, and the porous-medium region Ω_p is governed by the Darcy equation. The two regions are coupled on the interface Γ by the normal-flux condition, the normal-stress condition, and the Beavers–Joseph–Saffman condition.

Unless otherwise stated, the parameters are set as $g = 1$, $\gamma = 0.1$, $\varepsilon_{\mathrm{obj}} = 10^{-5}$, and the body force terms are neglected. For the free-flow region, a prescribed inlet velocity is imposed on $\Gamma_{f,D}$, and a natural boundary condition $T_f n_{\Omega_f} = 0$ is imposed on the outlet boundary $\Gamma_{f,N}$. For the porous-medium region, a fixed hydraulic head $\phi = 0$ is imposed on $\Gamma_{p,D}$, and no-flux conditions are imposed on the remaining porous boundaries.

The interface is updated by a moving body-fitted mesh. The mesh connectivity is kept fixed during the optimization. The FreeFEM++ function `checkmovemesh` is used to check mesh validity. If the deformed mesh does not satisfy the mesh-quality requirement, the step size is repeatedly halved until an admissible mesh is obtained. No remeshing is used in the present codes.

The two-dimensional examples use the volume-type Eulerian derivative, while the three-dimensional examples use the interface-type Eulerian derivative. In Case 4.1.2, both the distributed and interface-type Eulerian derivatives are also validated by a Taylor test.

3 Sec 4.1 Single-interface Around a Porous Region

This folder contains the codes for Example 1. It considers a Stokes–Darcy coupled system with a single moving interface around a porous region. In this example, the velocity Dirichlet boundary is divided into the inflow boundary $\Gamma_{f,i}$ and the no-slip wall boundary $\Gamma_{f,w}$. The prescribed inflow profile $\boldsymbol{u}_d$ is imposed on $\Gamma_{f,i}$, and homogeneous no-slip conditions are imposed on $\Gamma_{f,w}$.

Case 4.1.1

Case 4.1.1 is motivated by surface water flow over a permeable riverbed sediment layer. The computational domain is $\Omega = [0,2]\times[0,1]$, where an elliptical porous-medium region is embedded in the lower part of the domain:

$$\Omega_p = \left\{ (x,y) \in \mathbb{R}^2 : \frac{(x-1)^2}{0.5^2} + \frac{y^2}{0.4^2} \le 1,\ y \ge 0 \right\}.$$

The parameters are $\beta_1 = 1000$, $\beta_2 = 1.0$, $V_0 = 1.1|\Omega_f^0|$, $\mu_f = 1$, $k_p = 10^{-4}$.

The code solves the state and adjoint systems, computes the shape gradient, and updates the fluid–porous interface by the moving-mesh method. It also evaluates the mass-conservation residual in the Stokes region and the residuals of the Stokes–Darcy interface conditions, including the normal-flux condition, the normal-stress balance, and the Beavers–Joseph–Saffman condition. This case shows that the interface evolves into a smoother shape, the energy dissipation decreases, and the coupling conditions are accurately enforced while the volume constraint is maintained.

Case 4.1.2

Case 4.1.2 considers another initial single-interface geometry with a more strongly curved fluid–porous interface. The parameters are $\mu_f = 1$, $k_p = 10^{-3}$, $\beta_1 = 100$, $\beta_2 = 0.01$, $V_0 = 1.02|\Omega_f^0|$. The code follows the same optimization loop as Case 4.1.1. This case is used to test the robustness of the proposed moving body-fitted mesh method for a more curved initial interface. It is also used in the manuscript to validate the derived Eulerian shape derivatives by a Taylor test.

4 Sec 4.2 Multi-interface Optimization in an L-shaped Configuration

This folder contains the code for Example 2. It considers a multi-interface shape optimization problem in a two-dimensional L-shaped configuration. The computational domain is $\Omega = [0,1]^2$. A uniform inlet velocity $\boldsymbol{u}_d = [1,0]^T$ is imposed. The main parameters are $\mu_f = 0.5$, $k_p = 5 \times 10^{-5}$, $\beta_1 = 150$, $\beta_2 = 10^{-3}$, $V_0 = 0.6|\Omega_f^0|$. Compared with Sec 4.1, this example contains multiple fluid–porous interface components and non-smooth geometric features. The code constructs the

L-shaped geometry, solves the Stokes–Darcy state and adjoint systems, computes the distributed Eulerian derivative, and moves all admissible interface components during the optimization process. A perimeter regularization term is included to smooth the interface evolution.

5 Sec 4.3 Diffuser Model

This folder contains the codes for Example 3, the diffuser model with porous walls. It includes both two-dimensional and three-dimensional implementations. This example investigates the influence of the porous-medium permeability on the optimized interface shape and flow performance. The viscosity is fixed as $\mu_f = 1.0$, and the permeability values are $k_p = 0.01$, $k_p = 0.0025$. For the permeability study, the target volume and optimization parameters are kept fixed in each dimensional setting, and only the permeability k_p is varied.

Two-dimensional code

For the two-dimensional diffuser case, a uniform inlet velocity $\boldsymbol{u}_d = [0.1, 0]^T$ is imposed. The optimization parameters are $\beta_1 = 30$, $\beta_2 = 10^{-4}$, $V_0 = 1.35|\Omega_f^0|$. The code solves the 2D Stokes–Darcy state and adjoint systems, computes the shape gradient, and updates the diffuser interface. It outputs the optimized shape, velocity field, pressure fields, objective history, and volume-error history.

Three-dimensional code

For the three-dimensional diffuser case, a uniform inlet velocity $\boldsymbol{u}_d = [0, 1, 0]^T$ is imposed. The parameters are $\beta_1 = 10^3$, $\beta_2 = 1$, $V_0 = 1.5|\Omega_f^0|$. The code extends the diffuser model to a 3D Stokes–Darcy coupled system. It uses the interface-type Eulerian derivative and demonstrates that the proposed shape-gradient method can be applied to three-dimensional interface optimization problems.

6 Sec 4.4 Embedded Vessel–Tissue Coupling Model

This folder contains the codes for Example 4, the embedded vessel–tissue coupling model. It includes both two-dimensional and three-dimensional implementations. The free-flow region represents a vessel-like channel, and the porous region represents the surrounding biological tissue.

Two-dimensional code

The two-dimensional code considers a planar vessel–tissue configuration and studies the influence of the fluid viscosity on the optimized interface. The viscosity values are $\mu_f = 1.0$, 0.1, 0.01. The code solves the coupled state system, computes the adjoint variables and shape gradient, updates the vessel–tissue interface, and records the optimized flow fields and convergence histories.

Three-dimensional code

The three-dimensional code considers a spatial embedded vessel–tissue configuration. In the reported 3D test, the viscosity is $\mu_f = 0.01$. The code solves the 3D Stokes–Darcy system, computes the interface-type Eulerian derivative, updates the vessel–tissue interface, and outputs the optimized geometry, velocity field, pressure fields, objective history, and volume-error history.